GORENSTEIN RINGS, INVERTIBILITY OF THE DUALIZING COMPLEX AND OPENNESS OF THE GORENSTEIN LOCUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The localization of a dualizing complex at a prime is a dualizing complex over the local ring. Invoked inside the counted proof (source lines 4022-4024). Transports the dualizing complex to every local ring for the local statements.
- The uniqueness of dualizing complexes up to an invertible object. Invoked inside the counted proof (source lines 4024-4026). Turns the local Gorenstein condition into invertibility of the localized complex.
- The locality of the property of being a dualizing complex. Invoked at the end of the counted proof (source lines 4057-4058). Lifts the local invertibility to the global statement in the second part.

Source reference. Original lemma lemma-dualizing-glue: sources/stacks-dualizing/pinned-original-source.tex, source lines 2977–2977. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing-glue, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-dualizing-localize: sources/stacks-dualizing/pinned-original-source.tex, source lines 2950–2950. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing-localize, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-dualizing-unique: sources/stacks-dualizing/pinned-original-source.tex, source lines 2919–2919. Navigation pointer only: the counted

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Source reference. Original reference more-algebra-lemma-invertible-derived: sources/stacks-dualizing/pinned-original-source.tex, source lines 2852–2852. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-invertible-derived, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original section section-dualizing: sources/stacks-dualizing/pinned-original-source.tex, source lines 2768–2768. Navigation pointer only: the counted text refers to the Stacks item section-dualizing, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring.*

(1) *If A has a dualizing complex $\omega_A^\bullet$, then*
   (a) *A is Gorenstein $\Leftrightarrow$ $\omega_A^\bullet$ is an invertible object of $D(A)$,*
   (b) *$A_\mathfrak{p}$ is Gorenstein $\Leftrightarrow$ $(\omega_A^\bullet)_\mathfrak{p}$ is an invertible object of $D(A_\mathfrak{p})$,*
   (c) *$\{\mathfrak{p} \in \mathrm{Spec}(A) \mid A_\mathfrak{p}$ is Gorenstein$\}$ is an open subset.*

(2) *If A is Gorenstein, then A has a dualizing complex if and only if $A[0]$ is a dualizing complex.*

Proof. For invertible objects of $D(A)$, see More on Algebra, Lemma more-algebra-lemma-invertible-derived and the discussion in Section section-dualizing.

By Lemma lemma-dualizing-localize for every $\mathfrak{p}$ the complex $(\omega_A^\bullet)_\mathfrak{p}$ is a dualizing complex over $A_\mathfrak{p}$. By definition and uniqueness of dualizing complexes (Lemma lemma-dualizing-unique) we see that (1)(b) holds.

To see (1)(c) assume that $A_\mathfrak{p}$ is Gorenstein. Let n_x be the unique integer such that $H^{n_x}((\omega_A^\bullet)_\mathfrak{p})$ is nonzero and isomorphic to $A_\mathfrak{p}$. Since $\omega_A^\bullet$ is in $D^b_{Coh}(A)$ there are finitely many nonzero finite A-modules $H^i(\omega_A^\bullet)$. Thus there exists some $f \in A$, $f \notin \mathfrak{p}$ such that only $H^{n_x}((\omega_A^\bullet)_f)$ is nonzero and generated by 1 element over A_f. Since dualizing complexes are faithful (by definition) we conclude that $A_f \cong H^{n_x}((\omega_A^\bullet)_f)$. In this way we see that $A_\mathfrak{q}$ is Gorenstein for every $\mathfrak{q} \in D(f)$. This proves that the set in (1)(c) is open.

Proof of (1)(a). The implication $\Leftarrow$ follows from (1)(b). The implication $\Rightarrow$ follows from the discussion in the previous paragraph, where we showed that if $A_\mathfrak{p}$ is Gorenstein, then for some $f \in A$, $f \notin \mathfrak{p}$ the complex $(\omega_A^\bullet)_f$ has only one nonzero cohomology module which is invertible.

If $A[0]$ is a dualizing complex then A is Gorenstein by part (1). Conversely, we see that part (1) shows that $\omega_A^\bullet$ is locally isomorphic to a shift of A. Since being a dualizing complex is local (Lemma lemma-dualizing-glue) the result is clear. □